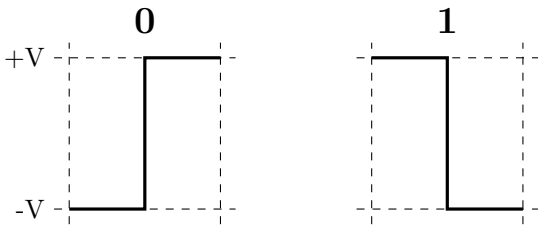
Cosc264: My Test Summery Sheet

Thomas Booker 56118260

Compiled on 14th September, 2026 at 12:32 Noon

Test on the 14th September (Week 8) at 6:30pm in JE233

Manchester

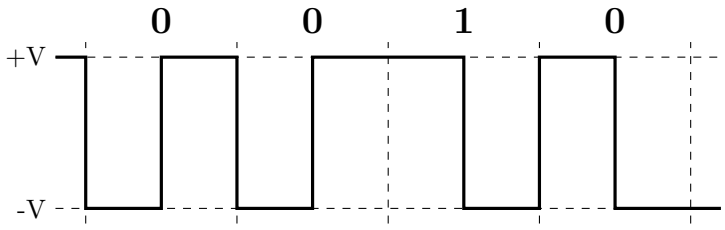


Differential Manchester

In Differential Manchester, the signal always flips at t=0.5.

If a 1 is encountered in the digital data, no signal flip is generated at t=0.

If a 0 is encountered in the digital data, a signal flip is generated at t=0.



Analog Carrier + Digital Data

$$s(t) = A(t) \cdot \cos\left((f_c + f(t)) \cdot t + \phi(t)\right)$$

s	Signal
A	Amplitude
f_c	Center frequency
f	Frequency offset
ϕ	Phase (wave offset)

PSK Phase-shift keying

$$s(t) = A \cdot \cos\left(f_c \cdot t + \phi(t)\right)$$

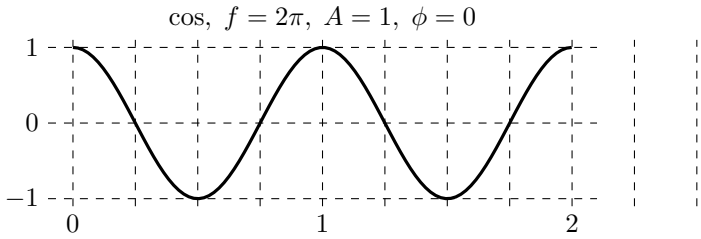
FSK Frequency-shift keying

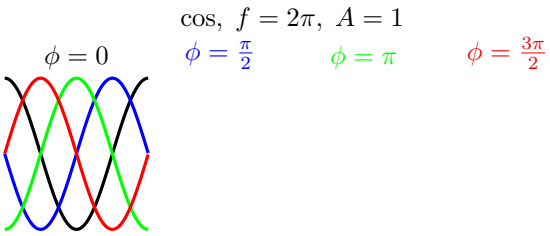
$$s(t) = A \cdot \cos\left((f_c + f(t)) \cdot t\right)$$

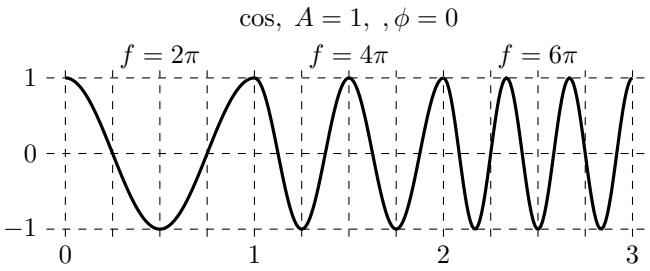
ASK Amplitude-shift keying

$$s(t) = A(t) \cdot \cos\left(f_c \cdot t\right)$$

Example Diagrams







Medium Access Control (MAC)

Broadcast/Shared medium, transmission are heard by all stations. If two transmissions overlap, both are garbled (Not true in some system we don't talk about).

Orthogonal Schemes: Split a chancel into smaller sub-chancel that can be transitioned on without collation between the sub-chancels.

The **medium access delay** is the time a station must wait to start transmitting (access the medium).

Frequency Division Multiple Access (FDMA)

The given channel bandwidth is subdivided into N sub-channels. Guard bands: reduction of adjacent-channel interference, robustness against imperfect frequency synchronisation. Access Delay is 0.

Time Division Multiple Access (TDMA)

Time is subdivided into superframes of duration T_{SF}. Each superframe is subdivided into N time-slots. There are short guard times between time slots.

Neglecting guard times, each sub-chancel gets the full channel bandwidth for a fraction $\frac{1}{N}$ of the time. So, the bit rate for a sub-chancel is $\frac{B}{N}$. The average medium access delay for one sub-chancel is $E\left[\text{Access Delay}\right] = \frac{T_{SF}}{2}$.

Random Access Protocols

Random access protocols:
Do not attempt to reserve channel resources for long times.
Do not require a central station or (much) shared state.
Do not access the medium at predictable times.
Often have low complexity (e.g. little coordination, if any).
Typically involve some random element.

Random access protocols accept risk of collisions to save coordination overhead and have overall improved efficiency! How well this trade-off works depends for example on the offered traffic load.

Pure ALOHA

Uses checksum to drop garbled packets. All packets are transmitted immediately with no coordination. Upon transmission a acknowledgement timer is started. The receiver sends an immediate ack upon successful reception (without a collision or transmission error); If not successful, the receiver remains quiet. If the transmitter receives an ack before the ack timeout, the frame is forgotten and the ack timer is cancelled. If the ack timer expires, transmitter enters a backoff mode: A random amount of time is waited and then the packet is re-transmitted. If the number of failed re-transmission trials exceeds threshold, the frame is dropped. New packets ready for transmission will wait in a queue if the station is backlogged.

The vulnerability period is 2 times the frame length. When more stations are active the collision probability increases. ALOHA cannot distinguish between collisions and channel errors destroying a frame.

Carrier Sense Multiple Access (CSMA)

All stations can determine the state of the medium (almost) instantaneously. Called carrier-sensing (CS) or clear-channel assessment.

Listen-before-talk (LBT): Before station transmits a frame, it performs CS operation. If channel is busy, station defers transmission according to one of several possible strategies.

CSMA protocols do not eliminate collisions completely, but reduce their rate or their impact.

CSMA approach is especially useful when medium propagation time is small compared to packet length, since other stations notice transmission quickly after it started. Typically satisfied in LANs, propagation delay is small Collisions can only occur when two stations start transmitting at almost same time (time difference smaller than propagation delay).

When propagation time is large compared to packet length, the sender might have already stopped transmission when receiver senses an idle carrier for the first time. Example: multi-access satellite configurations. Here LBT is almost useless, ALOHA is reasonable.